

Improving the Fleet Monitoring Management, through a Software Platform with IoT

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Abstract—IoT can provide a stable support to the management of fleets, by means of the use of diverse sensors and the development of software tools that allow the storage and the treatment of the sensed data. Considering the advance of the technologies and the companies, the development of the software systems must support high availability, reliability, and scalability, being these attributes of optimum quality for the management of fleets. This article introduces an online system that incorporates IoT, microservices, containers and the treatment of the OBD (On-Board Diagnostics) for each vehicle in the fleet, with the ultimate goal of achieving effective monitoring of it. The results of the usability analysis of the platform are presented in conjunction with the road tests, as well as the conclusions.

Keywords—fleet management platform, IoT, OBD, web and mobile application

I. INTRODUCTION

Fleet management addresses transportation management and monitoring, including cars, trucks, airplanes, ships, among others [1], and it allows companies which rely on transportation to remove or minimize the risks associated with investing in vehicles as well as knowing the behavior of drivers. In the same way, vehicle maintenance procedures are essential, as well as identifying the location of the driver in real time. As such, knowing the last time the vehicle made the oil change, or if the Vehicle Technical Inspection (VTI) is about to expire, allows the managers and managers of the companies to have an integral control of the fleet.

Different disciplines have address solutions for the fleet management optimization, but at present time, the advancement of technologies has allowed the development of new paradigms and systems that approach an integral solution for their monitoring. Wireless sensors cross different aspects of daily life, and allow to measure, act and understand different problems based on the domain in which they are. The proliferation of these sensors in a communication network makes up the Internet of Things, commonly abbreviated as IoT, in which sensors and actuators adapt to the environment, and information is shared between platforms. It is feasible to understand IoT as a technological paradigm that is conceived as a global network of machines and devices, which are capable of interacting with each other [2].

In this context, IoT has abandoned its childhood thanks to the adaptation of an endless number of wireless technologies, such as RFID (radio-frequency identification) and sensor nodes, and is progressively becoming the technology that will transform the Internet [3]. One of the fruitful domains of IoT is fleet management, addressing several important issues for

companies that own vehicles such as safety, efficiency, routes logistics, remote diagnosis, real time monitoring, and preventive maintenance.

Several authors described the construction of components or hardware systems to achieve vehicle monitoring [4], while others have also proposed the development of software for the treatment and visualization of data [5]. Although the objectives from these tools and our tool are the same, the existing tools integrate different types of sensors and not always describe the software technologies for management.

The present article aims to introduce and describe the considerations and development of a fleet management software platform, with technologies and concepts of Cloud Computing, IoT, and OBDs; which includes a web and a mobile version. Also, the description and results of a usability test performed on the platform, along with road tests to monitor the OBD in operation will be described. The main goal of the software platform is that it allows the fleet manager to locate the vehicles in real time, and obtain accurate and timely data about the whereabouts of the vehicle, receiving alerts and notifications, visualizing and interacting with statistics of drivers or vehicles, seeking to combine a comprehensive platform for monitoring vehicles and evaluating driver behavior.

The article is structured as follows: Section 2 addresses the methodology that includes the problem formulation and state of the art, the scope of the platform, the design and implementation of the platform, and the test of the hypothesis through a usability test and road tests; while the Section 3 shows the results obtained from these tests. Finally, Section 4 describes the conclusions and future work.

II. METHODOLOGY

A. Problem formulation and state of art

Inappropriate management of the company fleets leads to inefficient routes, requiring a bigger amount of routes and vehicles; as well as problems to establish homogeneous routes which leads to drivers having a lower level of knowledge, and may even rise the number of accidents. There may even be problems with the scheduling of visits and delivery, increasing the kilometers traveled per route, the use of the vehicles and the costs of vehicle maintenance.

At the organizational level, fleet management addresses the logistics associated with the administration of vehicles, and can include a wide variety of functions such as maintenance, monitoring, diagnosis, driver management, fuel, mileage, safety, among others. Currently, managers realize

that they need the development of knowledge-based decision support systems, where modeling, algorithmic intelligence and optimization systems are an integral part of decision-making processes [6].

Several authors have approached different technologies to achieve fleet management fulfilling some of the IoT categories for business applications presented by Lee and Lee [2]: monitoring and control; big data and business analytics; and, information sharing and collaboration. Chadil et al., [7] propose a hardware device prototype to achieve GPS, GRPS and Google Earth monitoring; which although it has the objective of managing a fleet, the implementation of it leads to a fundamental difference with our work with the construction of a hardware to achieve that goal.

Then, Malekian et al., [8] exploit OBD II, GPS and WiFi for the development of a fleet management system, to measure speed, distance and fuel consumption. In this case, as a common point we have the use of OBDII, and in comparison, we receive data from other factors besides those already mentioned such as sudden accelerations and braking, disconnection, temperature, idle, shocks, and changes of fast lane. Aljaafreh et al., [9] propose a monitoring system that uses GPS, WiFi, OBD, RFID; and a web software for management and remote diagnosis. These authors developed an embedded system installed in the vehicle, and which reports the mileage, fuel consumption, the degree of mismatch, the assignment of the driver, and that allows to manage the problem of vehicle maintenance and the state of the vehicle driver.

Although there are several articles that raise the development of systems or platforms for fleet management, these can present errors or incorrect math such as the calculation of fuel, the presence of delays to show real-time information on the location of the vehicle, vehicles spend a lot of time in traffic jams, and not all include optimal maintenance alerts or expirations. The hypothesis of the present work is that the design and development of a management platform based on IoT and with current technologies, will allow a company to manage its vehicles comprehensively, both through an application web as mobile; managing real-time alerts; as way to solve the problems mentioned above. One of the characteristics of the platform presented here is that it advocates, through its architecture and infrastructure, to achieve quality attributes like interoperability, availability, elasticity, and security.

Consequently, having these points in mind, the following scheme was drawn: 1) to define the scope of the platform, 2) to design and implement the architecture of the platform, including an analysis of current technologies in conjunction with communication, registration and treatment of the data with the OBD; 3) to design and implement the graphic interface of the platform, with the modeling and visualization of graphics, alerts and reports on the platform; and 4) to test the hypothesis through an usability test and the platform with the OBD.

B. Scope of the platform

The scope can be specified as follows:

a) The control and quality in the conduction, in order to diminish the accelerations and abrupt stops by the drivers, and to monitor the idle; generating driver profiles,

- b) Mechanical diagnosis and maintainance, to collect data on mileage and fuel consumption,
- c) Completed routes monitoring, with start and end points,
- d) The monitoring of events associated with changes of status (disconnection, isolation), and
- e) The generation of graphs and trends of the analyzed values, by date range.

C. Design and implementation of the platform architecture

1) *Technology analysis*: Considering the scope, it was necessary to address a study of the technologies and forms of development and hosting of applications nowadays, in order to obtain a scalable and available system, in real time. Considering, on the one hand, that in 2017, a survey by Intel Security showed that the hybrid adoption of the cloud grew three times in the last year, increasing from 19% to 57% of the organizations surveyed; that 73% of the companies planned to move within a period of two years to a data center completely defined by software; and that 49% of companies are delaying the deployment of the cloud due to a cybersecurity skills gap [10]; and on the other hand, which has undergone various changes in terms of ways of developing and deploying applications, such as containers as a service [11] and microservices [12]. As such, it was decided to select cloud computing, OBD, containers and microservices.

2) *General architecture of the platform*: Fig. 1 shows the general architecture of the system, which consists of two main components where the first provides communication with the OBD devices, and the second consumes the information collected by the first to be able to display it in a useful way for the user.

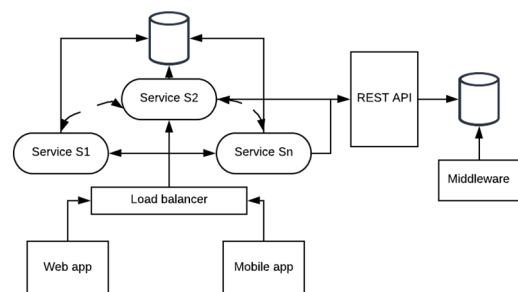


Fig. 1. General architecture of the platform.

As is it possible to observe, there are two applications working together, a web application that provides and contains views, session management, among others; and, the mobile application. The load balancer has the purpose of equalizing the load of the different services, and it is placed between the applications and the services. These services are those that deal with the communication with the database and the management of tokens to ensure the security of data. The web application and the mobile are nourished by the data provided by the database and update the views using the HTTPS protocol.

2) *Software system*: The software system for fleet management aims to allow the end user to monitor the fleet,

addressing not only the communication with the OBD devices, but also to consume the information that is collected to generate reports and useful notifications for the company and the drivers. This system was approached as a flexible and scalable web and mobile application, and Amazon Web Services (AWS, from now on) was chosen to host the application in the cloud because it is a platform that remains stable and on the rise [13]. When designing the architecture, it was decided to incorporate microservices as a strategy to achieve development agility, increasing the maintenance capacity and effective scalability of the application. In a way, microservices can be seen as the natural evolution of service-oriented architectures. In Fig. 2 (adapted from [14]), the architecture designed for the application can be seen.

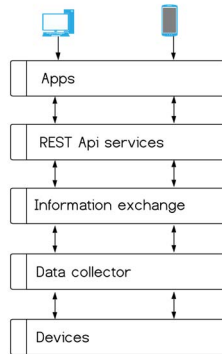


Fig. 2. Software architecture.

One of the central ideas of a microservices architecture is to be able to divide the functionalities in verticals through the implementation of a specific domain [15]. The infrastructure was implemented with Lagom and Docker, where Lagom is an open source framework that allows to build reactive microservices systems in Java or Scala, and which is based on Akka and Play. The services that are created in Lagom allow the concrete definition of the responsibilities in the development team, also obtaining systems with scalability, flexibility and elasticity; qualities necessary for the domain in which the system is embedded.

Later, Docker, the container platform, allows applications to be rapidly developed and executed, using a container virtualization platform [16]; whose main benefits are speed, consistency, density and flexibility. There are two main components: Docker Engine, which is responsible for creating and running containers; and Docker Hub, a cloud service to distribute containers [12]. Each Docker container is an isolated box that can contain the runtime environment for a particular service.

AWS is an ideal platform to implement Docker, because it offers a scalable infrastructure, API and SDK that are integrated into the development life cycle and accentuate the benefits of light and portable containers, offered by Docker. There are two ways to implement containers in AWS: on the one hand, AWS Elastic Beanstalk, which is useful for implementing a limited number of containers, or a web application; and on the other, Amazon ECS, which is specifically designed to allow you to run and manage containers through a series of hosts grouped in clusters [17]. For the architecture introduced in this paper, the second option has been chosen.

Given the nature of the project; Lagom, Scala, Cassandra, and Bootstrap were chosen for the web

application (Lagom, Scala, Cassandra, Bootstrap); Scala, Cassandra, and Google Maps API for the Rest-API; and Ionic2 for the mobile application. In this line, Cassandra was the chosen database, due to its scalability, availability, and its great ease of integration with AWS. For the web application, it was the Lagom framework that allowed the development of an architecture oriented to microservices, with high scalability; and the use of actors by both Lagom and Scala. Its biggest advantage is that it allows each request to be processed in parallel, allowing high concurrency and fast processing.

Then, the web application security was limited to those provided by the Play framework and the session encryption; and in the case of the Rest-API, through the use of tokens. As for the technologies to be used in the front-end in the web application, we chose the use of CoffeeScript, HTML, CSS and Bootstrap, to achieve a completely responsive website; along with the use of D3 as a data visualization library.

3) OBD and communication: One of the key components of the system is the On-Board Diagnostics (OBD), which was originally designed to reduce emissions by monitoring the vehicle's engine components. In this way, a basic OBD system consists of an electronic control unit that uses the input of sensors (such as oxygen) in order to control the actuators (like fuel injectors) in order to obtain the performance. There are two types of OBD: I and II, the OBDI is the first generation of OBD systems that were developed during the 1980s, and which used proprietary connectors, hardware interfaces and protocols. OBD-II is based on the standardization generated by the Society of Automotive Engineers, and ISO to describe the exchange of information between electronic control units and diagnostic tools [18].

In our context, the system architecture has the following elements: the OBDII [19], a GSM chip per vehicle and a remote server hosted in the cloud where the diagnostic codes registered by the OBD are stored in real time, establishing the position of the vehicle via GPS, and generating notifications about critical parameters. The registered parameters are: speed, RPM, mileage, consumption, accelerations and abrupt braking, temperature, disconnection, collisions and rapid lane changes. The transport of data packets via wireless channels between the server and the OBDII occurs through GPRS, the ETSI standard [20] as an extension of the GSM network for that purpose. Figure 3 shows the transmission protocol and safeguards the received data.

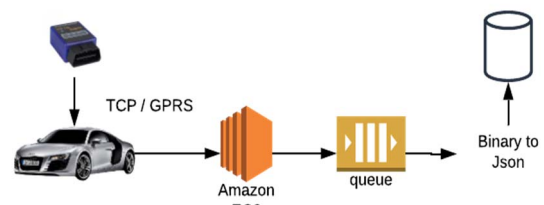


Fig. 3. Communication and data storage.

The data collector module has the task of establishing the connections with the multiple devices. For each of them, the various connectors have been developed, in which it is possible to exchange information through a permitted format: Json, XML, LCDP, MQTT, IP and LORA [21]. Following Fig. 2, the module that has the information exchange logic is the one that interacts and communicates with the data

collection module and the Rest-API. The latter is an interface between the systems that HTTP uses to obtain data and indicate the execution of operations on them. It is worth noting that within the Rest-API module, there are the different microservices and the load balancer. Finally, the applications module is responsible for the generation of a presentation layer for the analysis of information by the user.

D. Testing the hypothesis

In order to study the defined hypothesis, two types of tests were specified: on the one hand, a comprehensive study of the usability of the platform; and on the other, a road test with the use of vehicles and OBDs.

1) Usability test

Objectives: We aimed to conduct a usability study of the platform (web and mobile application), to determine the quality of the design through the evaluation of the platform by the end users.

Population: 40 users in the role of administrator and a moderator which explained the test goals.

Method: for the web version, the inspection methods of cognitive walkthroughs and feature inspection [22] were followed; while for the mobile version the Mobile Phone Usability Questionnaire (MPUQ) [23] was addressed.

Scheme for web version: the cognitive walkthroughs allow the evaluation the ease of learning in the design of an application, for which three general steps must be followed: (1) define the necessary data (users and evaluators, age range, previous knowledge), (2) go through with the actions (tour execution, evaluation), and (3) document the results (questions and answers). Consequently, within the step (1), the list of functionalities to be completed in the application was given to the sample (40 users in the role of administrator and a moderator), that included: create, read, update and delete (CRUD) operations of users, vehicles, drivers and fleets, interaction and visualization of the monitoring and real-time statistics of the vehicles in the fleet, visualizing graphics, maps and routes. In step (2), the questions were: A) are the actions available appropriately to the experience and knowledge of the user?, B) will users perceive that the correct action is available?, C) once the action has been found in the interface, will the users associate the action with the effect?, and D) once the action has been carried out, will the users understand the feedback of the application? Finally, in step (3) we proceeded to answer the questions in step 2 through a rating by Likert scale ("agree", "neither agree nor disagree", and "disagree").

Scheme for mobile version: the analysis of the six MPUQ criteria was carried out to achieve a study of (C1): ease of learning and use, (C2): assistance with operation and problem solving, (C3): emotional aspect and multimedia capabilities, (C4): commands and memory load, (C5): efficiency and control, and (C6): typical activities of a mobile phone.

2) Road test

Objectives: In order to evaluate the system designed and implemented, different functional and communication tests were carried out with the OBD.

Time: one (1) month

Population: 24 drivers with 23 vehicles and 23 OBDs; and 3 users (administrators).

Means of safeguarding data: spreadsheet that allowed to store different data such as vehicles, distances traveled, inconsistencies, errors in the displayed data, OBDs readings, among others.

Methodology: the drivers made trips of short (between 10 to 50 kilometers), medium (between 50 and 200 kilometers) and long distance (over 200 kilometers), in order to register the different points that were wanted to analyze: speed, revolutions per minute, mileage, consumption, accelerations and sudden stops, temperature, disconnection, collisions and changes of fast lane.

Scheme: On the one hand, the drivers were requested to achieve their regular routes. As such, they had to record the odometer's value before beginning each route, registering also the stops, and the fuel loads. On the other hand, the platform users were requested to monitor the performance of each driver and vehicle, in order to make a feasible comparison of the values shown in the platform.

III. RESULTS

A. Usability test

Addressing the analysis of step (3) of cognitive walkthroughs, it is possible to mention the following results. For question A), 87% of the answers were "agree" so that users could understand the set of tasks to be performed according to their experience and knowledge. Users highlighted: the monitoring and tracking of the vehicles in real time through maps generated by search schedules (with vehicle, driver, mileage, and date); and the download of reports for monthly and annual mileage, fuel consumption, and distances traveled, among others.

For question B), 69% of the answers were "agree", but it will be necessary to take measures and changes in the design to increase this percentage. It is feasible to mention that users highlighted the way in which the incidences of fleet vehicles are displayed (such as total mileage, accumulated mileage, consumption, average consumption, maximum speed, average speed, accelerations and sudden stops, disconnection, temperature, revolutions per minute, crashes, and fast lane changes); due to its presentation in the form of graphics and ordered by fleet and vehicle. In the same way, the functionality of the alerts for the next change of tires and expiration of insurance and VTI of the drivers were considered as extremely useful for the vehicle maintenance management.

Then, for question C), 66% of the answers were "agree", and the users mentioned with regard to the complete platform that the following points should be improved: the colors chosen; the traceability of steps in the functionalities (because, for example, to create a user, it was necessary to create a person before and then create the user for that person); to provide greater visual knowledge about what is being done by increasing the feedback to the user; and with respect to the reports generated from the platform, which although they are useful, the users pointed out that it might be needed a change on how the information is displayed and distributed in it.

Finally, question D) obtained 64% of "agree", and the users highlighted, for example, the quick feedback when interacting with various rankings and trends, both average distance and maximum speed reached by drivers, to analyze the behavior of drivers; together with the study of incidents in

particular together with issuance of notifications, both on the same platform and by e-mail in order to notify specific events or that require mediate or immediate attention: as disconnection or maximum speed exceeded.

As far as the mobile app is concerned, and considering MPUQ [23], the resulting values provided by the users who have interacted with the mobile application were schematized. In this way, from the percentages observed in Fig. 4 it can be inferred that the highest percentages were achieved in "typical tasks for mobile phones" (C6), and "commands and minimal memory load" (C4), and "efficiency and control" (C5). Which leads to the need for improvement in the points of "ease of learning and use" (C1), "assistance with operation and problem solving" (C2), and "emotional aspect and multimedia capabilities" (C3).

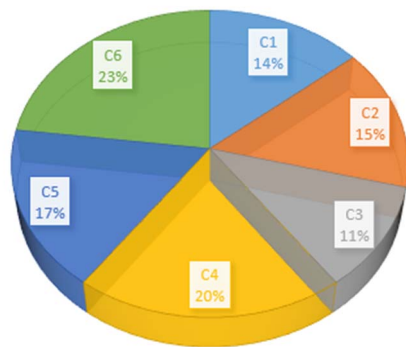


Fig. 4. Percentages of mobile analysis.

Based on the highlights, both for the web and mobile version, it is feasible to explain that it is necessary to address the improvement of ease of use such as user feedback and traceability. Consequently, a new iteration of the platform must include the points of improvement mentioned by the various users in order to increase and improve the usability level of said platform, both in its web and mobile version.

B. Test roads

As far as the management of the OBD is concerned, this has not presented any difficulty, but throughout different tests with different cars we have found that, based on the configuration of the same, it did not always record all the incidents that were needed (such as crashes or lane changes). Also, in certain trips not all detention or disconnection points were recorded; likewise it has happened that there were no recorded data for certain sections of some trips or routes (which can be seen in Fig. 5 when the lines of the cars are below the values according to the ranges of distances specified for short, medium and long distances).

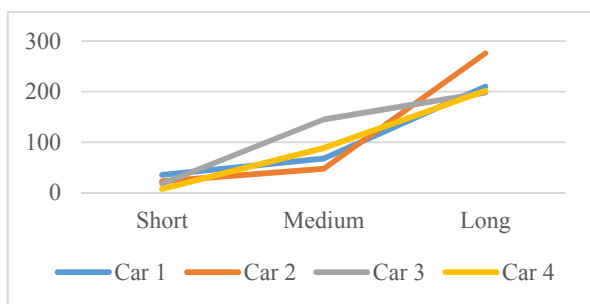


Fig. 5. Example of kilometers registered from four cars.

In the same way, rapid lane changes were not always recorded, which also had an impact on the analysis of driver behavior due to the lack of data for the complete study. Fortunately, as an advantage, the data that was received was stored correctly, and the system hosted in AWS supported the high availability and high concurrency of the various users. Also, the various notifications worked properly, which led to improved maintenance of the different parts of the vehicles, as well as the management of renewal and validity of the VTI, tires, insurance and registration of drivers.

IV. CONCLUSIONS

The various capabilities offered by the Internet of Things can save time and money for individuals and organizations, as well as help improve decision making and results in a wide range of application areas [24]. The Internet has drastically changed the way people live, and has impacted on the interactions between people in both work and personal contexts. The IoT has the potential to add a new dimension to this process by allowing communications with and between intelligent objects, leading to the "anytime, anywhere, anymedia, anything" communications vision [25].

The present work has introduced a system that allows the management of online fleets, which is a web and mobile platform hosted in AWS that, through communication by GPRS and TCP with the OBD installed in the vehicle, it is feasible to keep a record and real-time monitoring of the vehicles that make up the fleet.

In terms of lessons learned in the development of the software system, it is feasible to address first that having adopted the architectural style of the microservices allowed to have a common platform to develop, execute and administer the microservices, also allowing the adoption of the polyglot approach in what it refers to programming languages, storage, and operating systems [31]. It also contributed to the agile work of the team of developers with different levels of seniority, accelerating the independence of the team for the technologies. Microservices are justified by the context that requires flexibility in terms of scalability and availability.

Then, as far as containers are concerned, it was possible to implement service instances, and Docker helped in terms of automating the life cycle of system development. Regarding the infrastructure and architecture defined, it was possible to host an application in AWS that allowed to effectively control the fleet, through the reduction in service and maintenance costs, the generation of alerts or notification in real time of incidents in the vehicles or possible faults of parts of the same, the analysis panels, trends, maps and trends allow to efficiently perform the monitoring of the fleet, you can analyze the habits or behavior of drivers.

Consequently, and following the same line, instead of a vehicle receiving periodic maintenance, when using a system developed with sensors and IoT technology, it collects and analyzes the operation of the different parts of the vehicle, and sends that data to store them and perform an efficient treatment of them, including by fleet the state of the vehicles and drivers (in what refers to conduct of insurance and validity of the insurance, VTI and license).

Although the hypothesis was proven successful, its testing allowed to delimit the advantages, the points of improvement, the inconsistencies, and the future work. The usability test allowed us to define the improvement points in terms of user

experience, while the road test allows to test the platform in terms of consistency and communication with OBDs.

Currently, the platform is in a new stage of testing and analysis in order to correct errors. In the same way, increasing the number of vehicles in the fleet and the time of the tests. Also, a usability study of the tool is planned to analyze the perspective of users from a point of view of screens and interactions.

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